

Felipe Arbulú

Mathematical Engineer and PhD in Mathematics specialized in mathematical modeling with experience in quantitative modeling, data analysis, machine learning and scientific computing.

arbulu1f2008@gmail.com ✦ +33 6 68 00 82 03 ✦ Paris, France ✦ [LinkedIn](#) ✦ [GitHub Portfolio](#) ✦ [Personal web page](#)

WORK EXPERIENCE

Université de Picardie Jules Verne

Oct. 2020 – Present

Postdoctoral researcher Sept. 2025 – Present

Amiens, France

- Co-organizer of an international scientific conference; leading national and international research collaborations.

Attaché Temporaire d'Enseignement et de Recherche Jan. 2024 – Sept. 2025

Amiens, France

- Taught 384 hours in 6 courses, guiding multidisciplinary cohorts through weekly applied problem-solving sessions.

PhD Student Oct. 2020 – Dec. 2024

Amiens, France

- Developed novel mathematical models and analytical methods to solve complex problems, resulting in international collaborations, 10 conference presentations and 2 peer-reviewed publications.

Centro de Inteligencia Territorial (Universidad Adolfo Ibáñez)

Jun. 2017 – Jul. 2017

Centro de Investigación Avanzada en Educación (Universidad de Chile)

Jul. 2017 – Aug. 2017

Internships

Santiago, Chile

- Built geospatial data pipelines in Python (Pandas, NumPy), integrating territorial databases covering 22 major cities.
- Performed large-scale statistical analyses on socio-demographic datasets to support data-driven decision making.
- Collaborated with multidisciplinary teams to transform quantitative analyses into actionable policy recommendations.

Facultad de Ciencias Físicas y Matemáticas (FCFM) – Universidad de Chile

Mar. 2017 – Aug. 2018

Teaching Assistant

Santiago, Chile

- Advised graduate (Master and PhD) students in understanding complex mathematical concepts through exercise sessions, focusing on problem solving strategies and providing constructive feedback to enhance performance and engagement.

EDUCATION

Université de Picardie Jules Verne

Oct. 2020 – Dec. 2024

PhD in Mathematics

Amiens, France

Université Paris-Saclay

Sept. 2019 – Aug. 2020

Master of Science in Mathematics

Paris, France

Universidad de Chile

Aug. 2017 – Nov. 2018

Master of Science in Engineering, mention Mathematics

Santiago, Chile

Universidad de Chile

Mar. 2012 – Nov. 2018

Civil Engineering, mention Mathematics

Santiago, Chile

- Summa Cum Laude*; Dean's List, Rank 1/19 of the class

Universidad de Chile

Mar. 2012 – Aug. 2017

Bachelor of Science in Engineering, mention Mathematics

Santiago, Chile

- Dean's List

SKILLS & LANGUAGES

- Analytical:** Mathematical Modeling; Statistics; Machine Learning; Probability; Optimization; Numerical Analysis; PDEs.
- Programming & ML:** Python (Matplotlib, NumPy, Pandas, PyTorch, Scikit-learn); C++; MATLAB; Git; SageMath.
- Languages:** English (fluent); French (proficient); Spanish (first language).

ACADEMIC PUBLICATIONS AND PREPRINTS

- Topological weak mixing for certain linear involutions of hyperelliptic type.** Arbulú, Felipe, Espinoza, Bastián and Maass, Alejandro. 2026. [arXiv: 2607.16830](#).
- The Jacobs–Keane theorem from the S-adic viewpoint.** Arbulú, Felipe, Durand, Fabien and Espinoza, Bastián. 2024. Discrete Contin. Dyn. Syst. pp. 3077–3108. [doi: 10.3934/dcds.2024052](#). [arXiv: 2307.10663](#).
- Dynamical properties of minimal Ferenczi subshifts.** Arbulú, Felipe and Durand, Fabien. 2023. Ergodic Theory Dynam. Systems. pp. 3923–3970. [doi: 10.1017/etds.2023.7](#). [arXiv: 2207.14097](#).